

Governors

by

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Gravity Loaded or Dead weight Governor:

- The Porter governor is a modification of a Watt's governor, with **central load attached to the sleeve as shown in Fig. 1.**
- The **load moves up and down the central spindle.**
- This **additional downward force** increases the speed of revolution required to **enable the balls to rise to any predetermined level.**

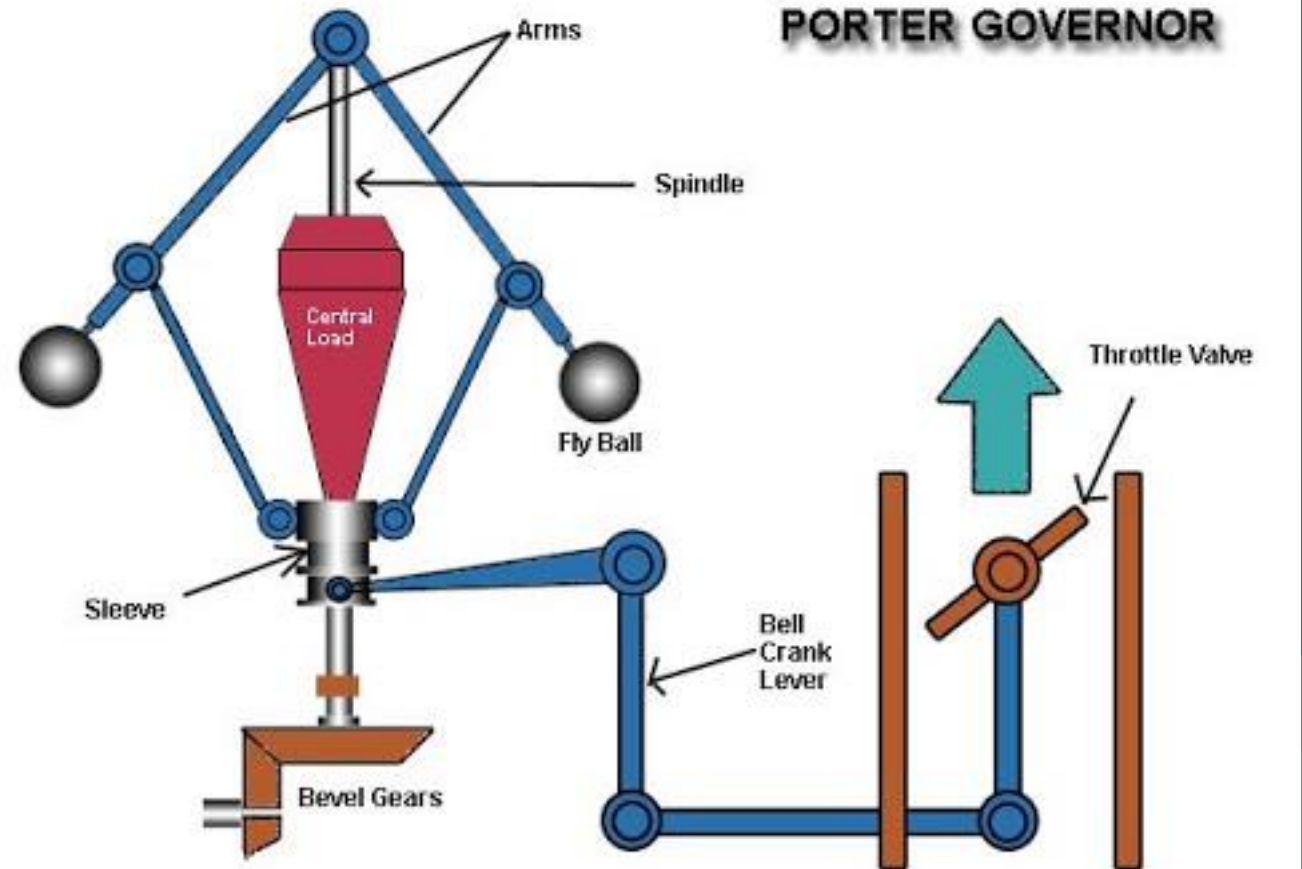


Fig. 1. Porter Governor



Gravity Loaded: Porter Governor contd...

m = Mass of **each ball** in kg

w = Weight of **each ball** in N

M = **Mass of the central load** in kg

W = **Weight of the central load** in N

r = **Radius of rotation** in m

h = **Height of governor** in m

N = **Speed of the balls** in rpm.

ω = Angular speed of the balls $(2\pi N/60)$ rad/s

F_C = **Centrifugal force acting on the ball** N

T_1 = **Force in the arm** in N

T_2 = **Force in the link** in N

α = **Angle of inclination of the arm to the vertical**

β = **Angle of inclination of the link to the vertical**

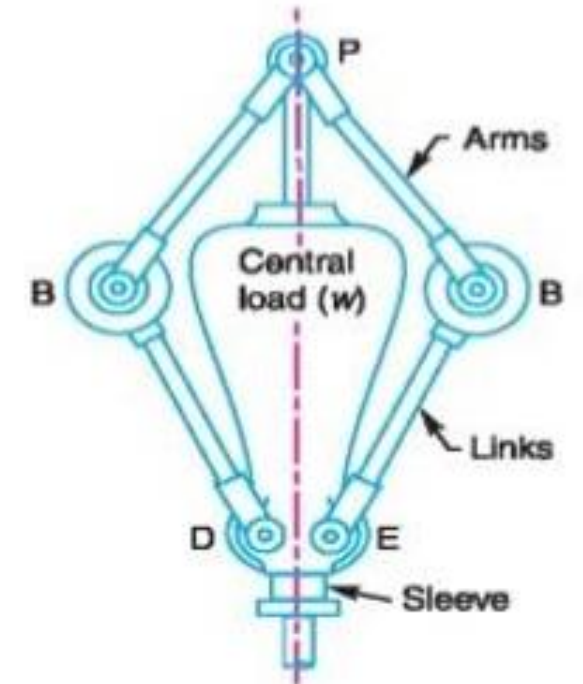
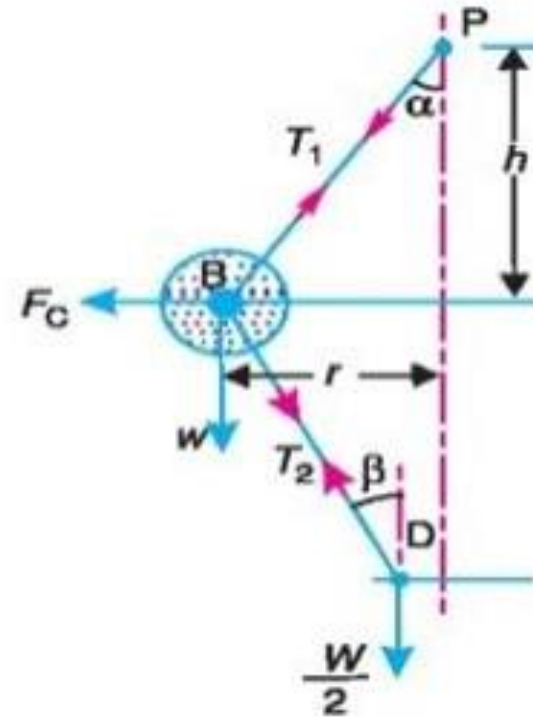


Fig. 2. Line diagram of Porter Governor



Porter Governor contd...

Method of resolution of forces

The equilibrium of the forces acting on point B is in equilibrium under the action of the following forces:

- (i) The weight of ball (w)
- (ii) The centrifugal force (F_c)
- (iii) The tension in the arm (T_1)
- (iv) The tension in the link (T_2)

Resolving the forces vertically ΣV :

$$T_1 \cos \alpha = T_2 \cos \beta + w$$

$$T_1 \cos \alpha = \frac{Mg}{2} + mg \quad (1)$$

$$\text{here, } T_2 \cos \beta = \frac{W}{2} = \frac{Mg}{2}$$

$$T_2 = \frac{Mg}{2 \cos \beta}$$

Resolving the forces horizontally ΣH :

$$T_1 \sin \alpha + T_2 \sin \beta = F_c$$

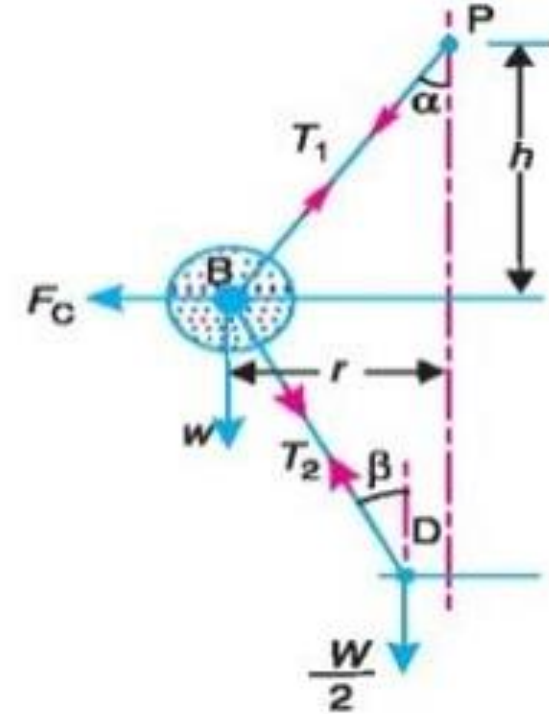


Fig. 3. Line diagram



Porter Governor contd...

$$T_1 \sin \alpha + \frac{Mg}{2 \cos \beta} \sin \beta = F_c$$

$$T_1 \sin \alpha = F_c - \frac{Mg}{2} \tan \beta \quad (2)$$

Dividing equation (2) by equation (1):

$$\frac{T_1 \sin \alpha}{T_1 \cos \alpha} = \frac{F_c - \frac{Mg}{2} \tan \beta}{\frac{Mg}{2} + mg} \gg \left(\frac{Mg}{2} + mg \right) \tan \alpha = F_c - \frac{Mg}{2} \tan \beta$$

$$\gg \left(\frac{Mg}{2} + mg \right) = \frac{F_c}{\tan \alpha} - \frac{Mg}{2} \frac{\tan \beta}{\tan \alpha}$$

Substituting, $\tan \alpha = r/h$ and $\frac{\tan \beta}{\tan \alpha} = q$

$$\gg \left(\frac{Mg}{2} + mg \right) = F_c \times \frac{h}{r} - \frac{Mg}{2} \times q$$



Porter Governor contd...

$$\left(\frac{Mg}{2} + mg \right) = m\omega^2 r \times \frac{h}{r} - \frac{Mg}{2} \times q$$

$$h = \left[mg + \frac{Mg}{2}(1+q) \right] \frac{1}{m\omega^2} \quad \text{here,} \quad \omega = 2\pi N/60 \text{ rad / second}$$

$$h = \left[\frac{m + \frac{M}{2}(1+q)}{m} \right] \frac{895}{N^2} \quad (3)$$

When **the length of arms are equal to the length of links** and the points P and D lie on the same vertical line, then $\tan\alpha = \tan\beta$ and $q = 1$

$$h = \left[\frac{m + M}{m} \right] \frac{895}{N^2} \quad (4)$$



Porter Governor contd...

When the loaded sleeve moves up and down the spindle; the frictional force acts on it in a direction opposite to that of the motion of sleeve.

$$h = \left[\frac{m + \frac{M \pm F}{2} (1 + q)}{m} \right] \frac{895}{N^2} \quad (5)$$

The + sign is used when the sleeve moves upwards or the governor speed increases and - sign is used when the sleeve moves downwards or the governor speed decreases.



Problem 1. A Porter governor has equal arms each 250 mm long and pivoted on the axis of rotation. Each ball has a mass of 5 kg and the mass of the central load on the sleeve is 15 kg. The radius of rotation of the ball is 150 mm when the governor begins to lift and 200 mm when the governor is at maximum speed. Find the minimum and maximum speeds and range of speed of the governor.

Solution. Given : $BP = BD = 250 \text{ mm} = 0.25 \text{ m}$;
 $m = 5 \text{ kg}$; $M = 15 \text{ kg}$; $r_1 = 150 \text{ mm} = 0.15 \text{ m}$;
 $r_2 = 200 \text{ mm} = 0.2 \text{ m}$

The minimum and maximum positions of the governor are shown in Fig. 4 (a) and (b), respectively.

Minimum speed:

when $r_1 = BG = 0.15 \text{ m}$,

$N_1 =$ Minimum speed Referring Fig. 4(a)

height of the governor:

$$h_1 = PG = \sqrt{PB^2 - BG^2} = \sqrt{0.25^2 - 0.15^2} = 0.2 \text{ m}$$

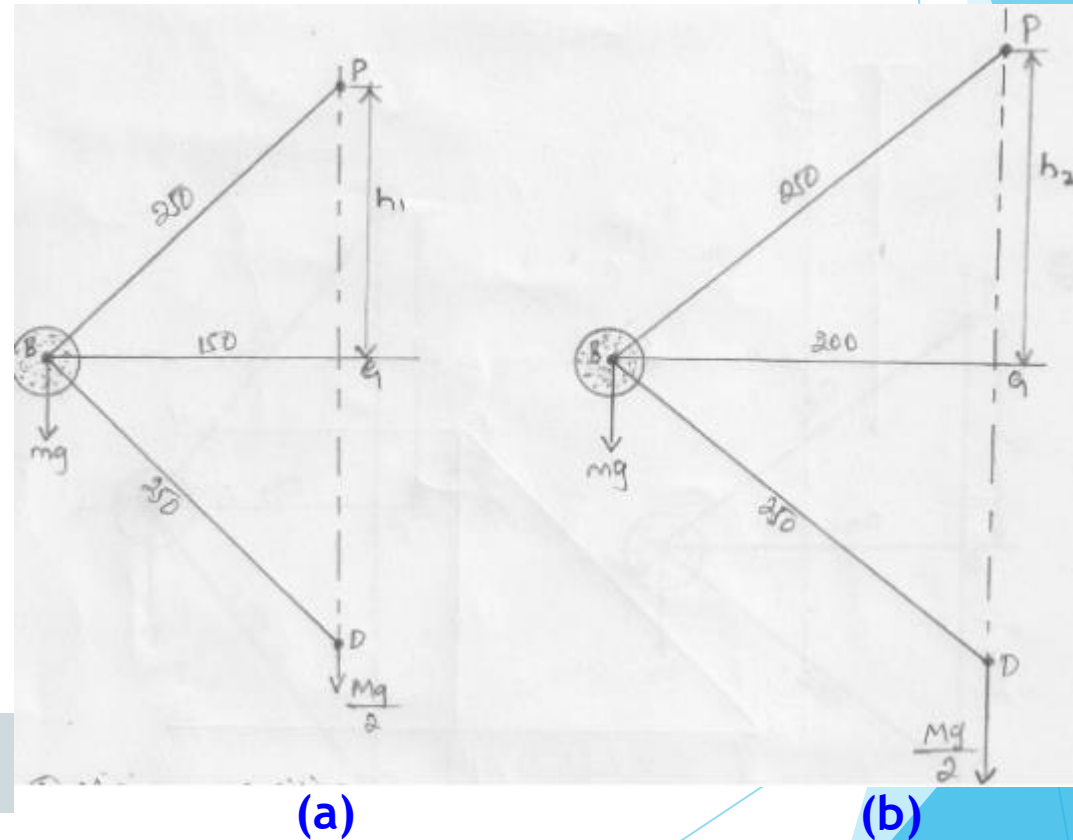


Fig. 4.



Problem 1. contd...

Minimum speed of the governor is given by:

$$N_1^2 = \left(\frac{m + M}{m} \right) \times \frac{895}{h_1} = \left(\frac{5 + 15}{5} \right) \times \frac{895}{0.2} = 17900$$

$$N_1 = 133.8 \text{ r.p.m.}$$

$$\text{Range of Speed } N_2 - N_1 = 154.5 - 133.8 = 20.8 \text{ rpm}$$

Maximum speed:

when $r_2 = BG = 0.2 \text{ m}$,

$N_2 =$ Maximum speed Referring Fig. 4(a)

height of the governor:

$$h_2 = PG = \sqrt{PB^2 - BG^2} = \sqrt{0.25^2 - 0.2^2} = 0.15 \text{ m}$$

Maximum speed of the governor is given by:

$$N_2^2 = \left(\frac{m + M}{m} \right) \times \frac{895}{h_2} = \left(\frac{5 + 15}{5} \right) \times \frac{895}{0.15} = 23867$$

$$N_2 = 154.5 \text{ r.p.m.}$$



Thank You

